Week 4 Tutorial Questions

Question 1: Explain in detail the key components of the State design pattern, using a real-world example to illustrate your explanation.

Question 2: A vending machine has various states like "NoSelection", "HasSelection", "InsertedMoney", and "Dispensed". The operations (events) a vending machine can perform are selectItem(), insertMoney(), dispenseItem(), cancel(), and refundMoney().

The behavior of these operations changes depending upon the current state of the machine. For instance, you can't dispense an item when there is "NoSelection", or you can't make a selection when the machine is already in the "Dispensed" state. Which is the best design pattern for the vending machine problem? Justify your answer. Draw a class diagram for the design of the system.

Question 3: A web server processes incoming HTTP requests through a series of operations, including authentication, logging, and data validation, before the final request is handled. Explain how the Chain of Responsibility pattern can be used to decouple these operations. In your answer, list the key components involved in this design.

Question 4: The Strategy Design Pattern is often used to implement the Open/Closed Principle (OCP). Using an e-commerce payment system as an example (e.g., Credit Card, PayPal, Bitcoin), discuss how this pattern adheres to OCP and provide two benefits of using this approach in a professional software environment.

Question 5: An e-commerce platform usually provides multiple delivery strategies for customers such as Express Delivery, Standard Delivery, Drone Delivery, In-Store Pickup and more. Each delivery type has its own set of rules and calculations for cost, delivery date, etc.

1. `Express Delivery`: This could involve a higher cost, but the item will be delivered the same day or the next day.
2. `Standard Delivery`: This typically has a lower cost, but the item might take several days to be delivered.
3. `Drone Delivery`: This involves a high cost, but offers speedy delivery.
4. `In-Store Pickup`: The customer will come to the store to pick up the item, so no delivery charge is applied.

Include all the delivery strategies into one delivery class can clutter the code and lead to challenges for code maintenance. Which design pattern is the best candidate to solve the problem? Justify your answer. Draw a class diagram for the design of the system.

Question 6: Suggest the scenario when Decorator pattern is appropriate and give a real-world example to illustrate your suggestion.